7/3/26

Test -2

1) What is HTML? Why is it used in web development?

HTML (HyperText Markup Language) is the standard language used to create and structure web pages.

It is used in web development to:

* Structure content on a webpage
* Display text, images, videos, links, etc.
* Create the basic layout of a website.

Example:

```
<! DOCTYPE Html>
<htm>
  <head>
    <title> My Page </title>
  </head>
  <body>
    <P> My website </P>
  </body>
  </html>
```

2) What is the purpose of <! DOCTYPE html>? Why is it important?

<! DOCTYPE html> tells the browser that the document is written in HTML 5.

Importance:

* Ensure the browser displays the page correctly
* Prevents compatibility problems
* Helps browsers follow modern HTML standards

3) D/B tags and Elements

Tags	Elements
* Keywords enclosed in angle brackets	* Complete structure opening tag, content and closing tag
* used to define HTML structure.	* Combination of tags + content.

Example

  < > → angle bracket
 Element

4) Semantic Elements:

Semantic Elements clearly describe their meaning to the browser and developer.

Examples : <header>
<section>
<footer>

```
<header>
    <h1> Hello </h1>
</header>
```

5) Non- Semantic Elements & why introduce in HTML5

Non semantic Elements do not describe their meaning.

Examples:

Introduced in HTML 5

* Improve website structure
* Better SEO
* Easier for developers to understand
* Better accessibility for screen readers

6) D/B and

* Makes text bold * Only visual styling Example: Hello	* Shows strong importance * Semantic meaning Example: Important message

7) Purpose of <a> tag & which attribute is required to create a hyperlink?

The <a> tag is used to create hyperlinks to another webpage or resource.

Required attribute: href

Example:

<a href = "https://example.com"> Visit website </a>

8) Four input types used in HTML forms:

Examples:

text, password, email, radio, checkbox, Submit

```
<input type = "text">
<input type = "email">
<input type = "radio">
<input type = "Submit">
```

9) Purpose of <video> and <audio> tags:

These tags are used to embed media files in a webpage.

<video> → for videos

<audio> → for sound/music

control attributes:

Add play, pause and volume buttons.

Example:

```
<video controls>
  <source src = "movie.mp4">
</video>
```

10) Purpose of Meta tags in HTML:

Meta tags provide information about the webpage to the browser and search engines.

Uses:

* character encoding
* SEO
* Page description
* viewport settings

Example:
<meta name = "description" content = "My website">

11) UTF-8 mean in <meta charset = "UTF-8">?

UTF-8 is a character encoding standard that supports almost all characters from different languages.

⇒ It allows the webpage to display:

- English
- Chinese
- special symbols

12) What is CSS? Why do we use CSS?

CSS (cascading style sheets) is used to style and design web pages.

CSS helps to:

* Change colors
* set fonts
* Content layout
* Make websites responsive

Example:

```
P{
    color: blue;
    font-size: 18px;
}
```

13) What are CSS selectors?

CSS selectors are used to select HTML elements to apply styles.

Three types:

1) Element selector
2) Class selector
3) ID selector

Element selector:

```
p{ color: red;
}
```

Class selector:

```
.box{
  background-color: yellow
}
```

ID selector:

```
#title {
  font-size: 30px;
}
```

14) CSS:

Cascading means multiple CSS rules can apply to the same element, and the browser decides which rule has priority.

Priority order:

1) Inline CSS
2) External CSS
3) Internal CSS

Example:

```
<p style="color: red;">
Hello</p>
```

15) CSS Box Model:

The CSS Box Model describes how elements are structured and spaced on a webpage.

It has 4 components

Content → padding → Border → Margin

```
<div class = "box"> Hello </div>

.box {
  width: 200px;
  padding: 20px;
  border: 5px solid black;
  margin: 30px
}
```